

ROTA AB, Spain

WMO: 084490

Lat: 36.639N

Lon: 6.332W

Elev: 26

StdP: 101.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 13025

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	4.4	-3.1	2.9	7.7	-0.9	3.5	8.1	12.9	13.1	11.2	13.0	3.4	40	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	33.8	21.1	32.0	20.7	30.2	20.4	23.6	28.7	22.8	27.6	22.2	27.0	6.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	16.8	25.8	21.1	15.8	25.0	20.8	15.5	24.7	70.4	28.8	67.5	27.7	65.0	27.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.7	8.6	0.9	37.0	1.8	1.7	-0.4	38.2	-1.5	39.2	-2.5	40.2	-3.8	41.4		
			-0.7	24.8	1.8	1.7	-2.0	26.0	-3.1	27.0	-4.1	28.0	-5.4	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	11.4	12.1	14.3	16.1	19.1	22.2	24.0	24.6	22.4	19.8	15.1	12.5	(6)
(7)		DBStd	5.25	2.52	2.53	2.32	2.35	2.72	2.60	2.43	2.48	2.36	2.44	2.84	2.65	(7)
(8)		HDD10.0	33	14	9	1	0	0	0	0	0	0	0	1	8	(8)
(9)		HDD18.3	908	216	174	124	74	23	1	0	0	1	13	101	182	(9)
(10)		CDD10.0	2892	57	68	136	184	281	366	433	452	372	305	155	84	(10)
(11)		CDD18.3	724	0	0	1	8	45	116	175	193	122	59	6	0	(11)
(12)		CDH23.3	5023	0	1	10	62	303	805	1341	1539	704	250	8	0	(12)
(13)		CDH26.7	1642	0	0	1	8	79	280	475	553	202	44	0	0	(13)
(14)	Wind	WSAvg	4.0	3.7	4.1	4.4	4.5	4.4	4.3	3.9	3.8	3.7	3.7	3.7	3.8	(14)
(15)	Precipitation	PrecAvg	536	71	61	55	43	27	8	0	2	21	75	93	78	(15)
(16)		PrecMax	908	232	274	259	121	99	54	3	32	85	243	361	341	(16)
(17)		PrecMin	261	0	0	0	0	0	0	0	0	0	0	1	2	(17)
(18)		PrecStd	168	53	52	50	30	27	12	1	5	23	65	78	73	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	21.2	24.2	27.9	31.2	34.8	35.2	35.9	33.1	30.0	24.3	20.2	(19)
(20)		MCWB	14.6	14.4	15.7	17.2	18.3	20.3	21.2	21.2	21.0	19.5	17.4	15.7	(20)	
(21)		2%	DB	17.9	19.1	22.2	25.1	28.9	31.9	33.2	33.8	31.1	27.9	22.4	19.0	(21)
(22)		MCWB	14.7	13.8	15.2	15.9	17.8	19.4	21.0	21.8	20.6	19.0	17.2	15.6	(22)	
(23)		5%	DB	17.0	17.9	20.9	23.0	26.8	29.9	31.2	31.9	29.1	26.0	21.1	18.0	(23)
(24)		MCWB	14.2	13.7	14.8	15.6	17.1	19.1	20.7	21.6	20.3	18.6	17.0	15.2	(24)	
(25)		10%	DB	16.0	16.8	19.0	21.0	24.6	27.9	29.1	29.8	27.0	24.2	20.0	17.1	(25)
(26)		MCWB	13.3	13.6	14.0	15.4	17.0	19.1	20.4	21.4	20.1	18.5	16.5	14.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	16.4	17.7	19.1	20.9	22.6	24.2	25.2	23.7	21.8	20.1	17.7	(27)
(28)		MADB	17.8	18.3	22.2	23.7	26.7	29.1	30.2	30.2	28.2	24.3	21.7	18.9	(28)	
(29)		2%	WB	15.6	15.5	16.4	17.7	19.6	21.5	23.2	23.9	22.6	20.8	18.7	16.7	(29)
(30)		MADB	17.0	17.3	19.8	21.6	24.9	26.8	28.4	29.0	26.9	23.9	20.9	17.8	(30)	
(31)		5%	WB	14.6	14.8	15.6	16.7	18.7	20.8	22.4	23.1	21.8	20.2	17.8	15.9	(31)
(32)		MADB	16.2	16.7	18.7	20.5	23.5	25.9	27.4	28.0	25.9	23.5	20.1	17.4	(32)	
(33)		10%	WB	13.6	13.9	15.0	16.0	18.0	20.2	21.8	22.4	21.1	19.6	16.9	15.0	(33)
(34)		MADB	15.5	16.1	17.8	19.5	22.4	25.3	26.7	27.3	24.9	22.9	19.2	16.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.3	8.4	8.5	8.1	8.4	8.4	8.6	8.7	8.2	8.3	8.0	7.9	(35)	
(36)		5% DB	MCDBR	8.9	10.2	11.9	12.5	12.7	12.8	12.6	12.1	11.8	11.3	9.2	8.4	(36)
(37)		MCWBR	6.4	6.6	7.1	6.8	6.3	5.7	5.7	5.6	5.6	5.7	5.7	5.8	(37)	
(38)		5% WB	MCDBR	6.7	7.9	9.2	9.2	9.7	9.6	9.7	9.8	8.4	8.3	7.1	6.2	(38)
(39)		MCWBR	5.6	6.0	6.3	6.0	5.7	5.1	5.2	5.3	4.8	4.9	5.2	5.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.353	0.392	0.388	0.401	0.414	0.401	0.419	0.406	0.383	0.351	0.337	(40)	
(41)		taud	2.503	2.447	2.326	2.362	2.353	2.333	2.360	2.320	2.379	2.445	2.498	2.504	(41)	
(42)		Ebn at Noon	860	881	872	894	883	868	876	854	847	835	829	828	(42)	
(43)		Edh at Noon	81	97	119	121	124	127	122	125	112	96	82	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.70	3.60	4.80	6.04	7.16	7.86	7.97	7.04	5.54	4.01	2.90	2.38	(44)	
(45)		RadStd	0.26	0.39	0.43	0.36	0.38	0.31	0.14	0.21	0.27	0.32	0.23	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62

Nomenclature: See separate page